Module 12 Lecture - Linear Regression and Correlation

Introduction to Statistical Methods

Quinton Quagliano, M.S., C.S.P

Department of Educational Psychology

Table of Contents

1 Overview and Introduction

1.1 Textbook Learning Objectives

- Discuss basic ideas of linear regression and correlation.
- Create and interpret a line of best fit.
- Calculate and interpret the correlation coefficient.

1.2 Instructor Learning Objectives

- Understand the concept of prediction and how it ties into linear regression
- Begin to see the broader applications of regression outside the simple applications

1.3 Introduction

- In this unit, we will be describing both ________ and ________ linear regression
  - At a basic level, these statistics are appropriate to ________ numeric variables with a ratio or interval scale of measurement
  - There are more advanced ________ that can work with ordinal and categorical data, but they are out of scope for this class

Discuss: As review, describe what it means for a variable to be continuous and what the different between ratio and interval scale data are?

- Furthermore, we'll be primarily focusing on the bivariate scenario, where there are ________ variables we are comparing
  - This is usually set up as one predictor variable and one criterion variable
  - There are important (and commonly) used extensions for multivariate scenarios, but they involve more complex interpretation and assumptions
- Your book uses the terms "independent" and "dependent" to describe predictor and criterion variables, but I stay away from those terms unless the ________ of the study was experimental in nature

Important

Linear regression often shows up in cross-sectional or longitudinal research where it is difficult to say it can fully support a causal claim. Be careful in describing what exactly a regression model shows about the relationship between two or more variables

2 Basic Review of Linear Equations

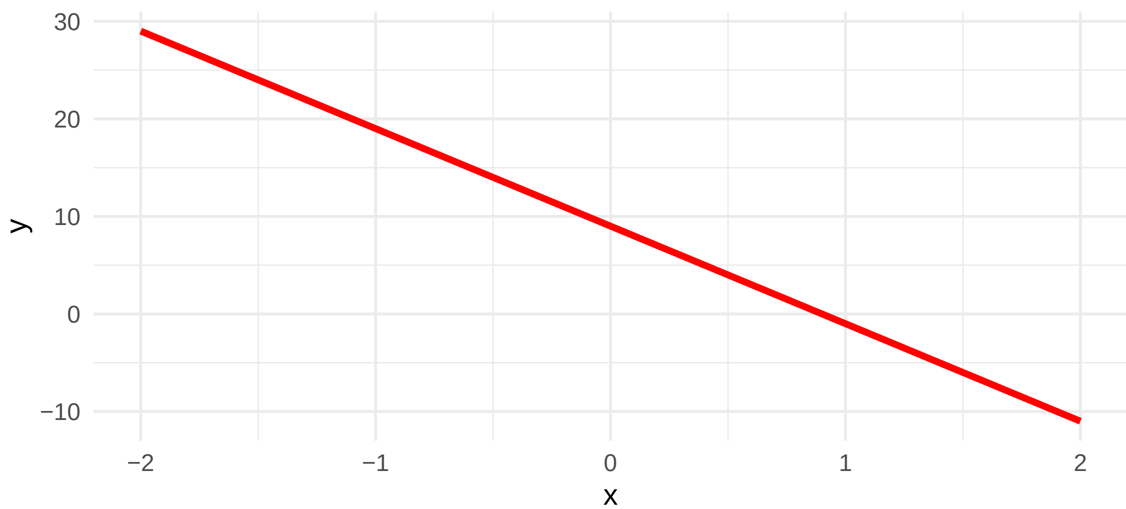
2.1 Introduction

- Many folks are introduced to the basic ______________ equation in algebra and/or geometry: $y = a + bx$
  - Where:
    - y = ______________ variable
    - a = ______________ or the value of y when $x = 0$
    - b = ______________
    - x = ______________ variable
  - ______________ forms include: $y = b + mx$ or $y = mx + b$
    - All of these formulas mean the ______________ thing
  - Signs of the slope and y-intercept can be ______________
- Examples:
  - $y = 8 + 4x$
  - $y = 12x - 3$
  - $y = -10x + 9$ - shown in the graph below

Listing 1 Line Plot of y = -10x + 9

```
x_vals <- seq(-2, 2, by = 0.1)
y_vals <- -10 * x_vals + 9 # NOTE: See equation here!
data <- data.frame(x = x_vals, y = y_vals)

ggplot(data, aes(x = x, y = y)) +
  geom_line(color = "red", size = 1.2) +
  labs(
    title = "",
    x = "x",
    y = "y"
  ) +
  theme_minimal()
```



Important

Pay attention to where the 0 on the x-axis is! It may not always be on the far left

- If put on a ______________ plot, this equation result in a straight line
  - If $b > 0 \rightarrow$ upward to right
  - If $b < 0 \rightarrow$ downward to right
  - If $b = 0 \rightarrow$ straight horizontal across
  - A ______________ a / y-intercept will result in the line crossing the y-axis ($x = 0$) at a higher point

? Looking at the 3 graphs above, which one has the lowest y-intercept

- (●) A) Left
- ( ) B) Center
- ( ) C) Right
- ( ) D) All have the same y-intercept

Explanation:

3 Scatterplots & Graphical Methods

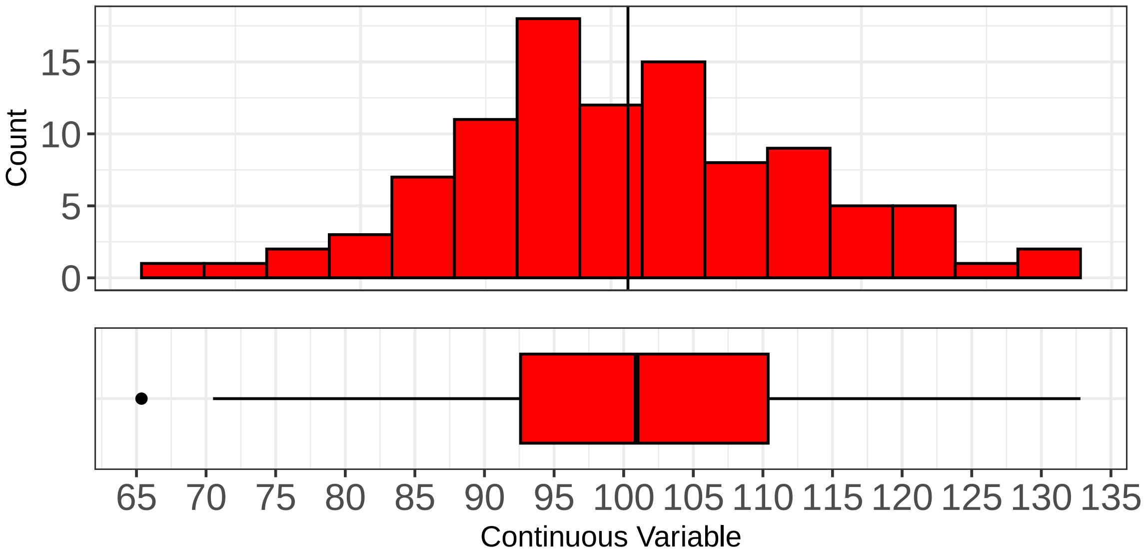
3.1 Introduction

- Early in the semester, we learned about graphical methods for showing ________ information about individual, continuous variables, i.e., boxplots, histograms, stem-and-leaf plots

Discuss: Review: Take a look at the histogram and boxplot below, what can you say to describe this variable based on what they plots tell you?

Listing 2 Histogram and boxplot of a hypothetical continuous variable

```
histbox(normal_data, value, "Continuous Variable")
```

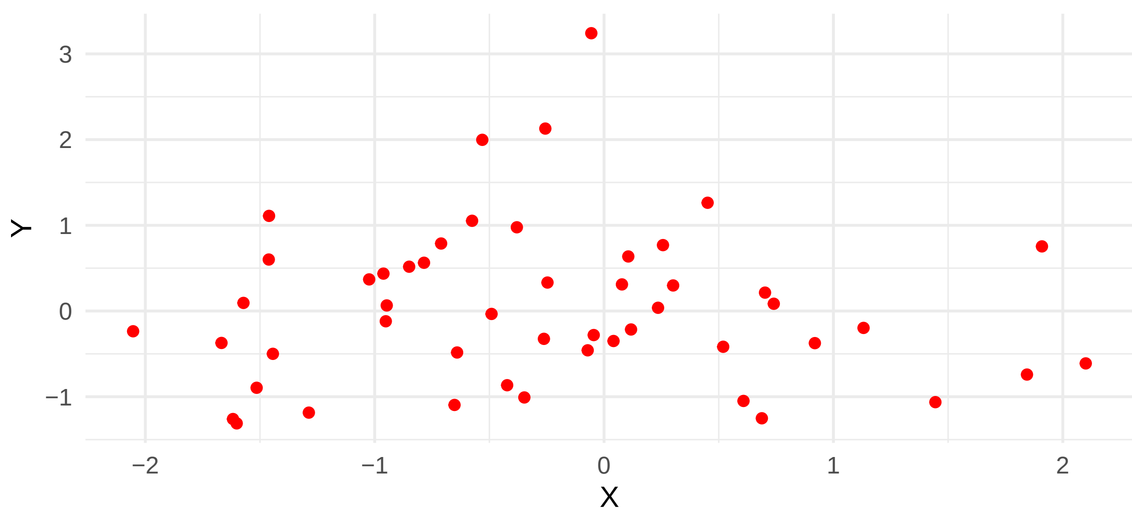


- In addition to ways we can graphically represent a single variable, we can also represent the ______________ scenario with a scatterplot
  - A scatterplot will be set up so that one of our ______________ is on the x-axis, and one of our variables is on the ______________
  - It's convention to put the ______________ variable on the x-axis, and the ______________ variable on the y-axis, but it's not necessarily "wrong" to do the other way

Listing 3 Scatterplot Example

```
# Create sample data
df <- data.frame(x = rnorm(50), y = rnorm(50))

# Create scatterplot with red points
ggplot(df, aes(x = x, y = y)) +
  geom_point(color = "red") +
  labs(title = "", x = "X", y = "Y") +
  theme_minimal()
```



> **Important**
>
> Remember, various graphs and plots are a great way for us to best understand our variables and relationships - we can and should use them often alongside our numeric results.

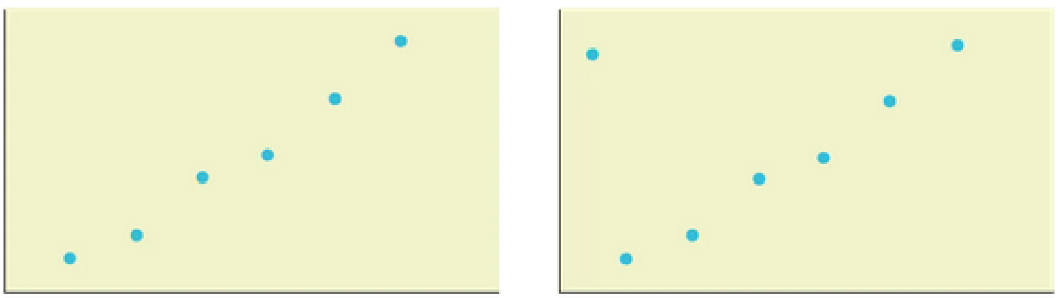
3.2 Strength and Direction

- In a scatterplot, we can __________ inspect the strength of the correlation, as well as the direction of the relationship
- The strength of a relationship indicates how closely __________ two variables are to one another
  - A __________ strength indicates close/strong relationship, and is visually shown by points on a scatterplot sitting along a clear linear trend

- Conversely, a ______________ strength would represent two variables not very related to one another, and be shown by a scatterplot where points do not seem to follow a line

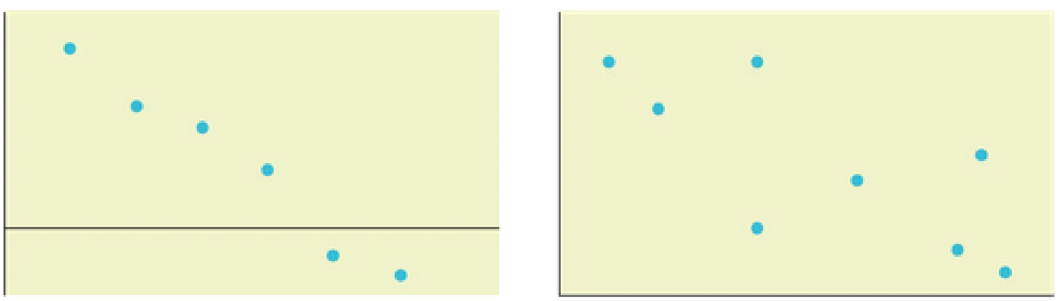
• The direction of a relationship indicates how one variable ______________ to change in the other

- A negative relationship indicates that as one variable increases, the other ______________
- A positive relationship indicated that as one variable ______________, so does the other



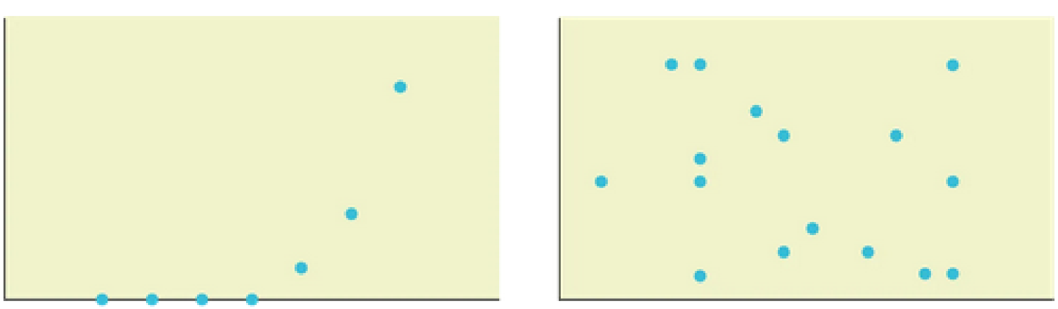
(a) Positive linear pattern (strong) (b) Linear pattern w/ one deviation

Figure 12.6



(a) Negative linear pattern (strong) (b) Negative linear pattern (weak)

Figure 12.7



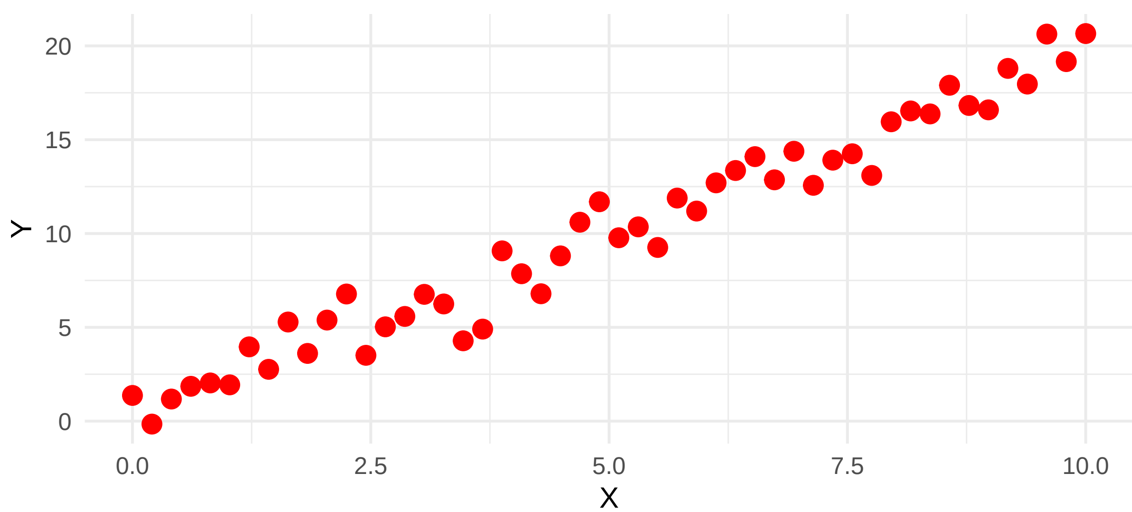
(a) Exponential growth pattern (b) No pattern

Discuss: Using the scatterplot below, describe the direction and strength of the relationship between the two variables x and y. Think about how an imaginary line between these points would look

Listing 4 Hypothetical Scatterplot

```
set.seed(42)
x <- seq(0, 10, length.out = 50)
y <- 2 * x + rnorm(50, mean = 0, sd = 1)
df <- data.frame(x = x, y = y)

# Create scatterplot
ggplot(df, aes(x = x, y = y)) +
  geom_point(color = "red", size = 3) +
  labs(title = "", x = "X", y = "Y") +
  theme_minimal()
```



4 Correlation Coefficient r

4.1 Introduction

- Pearson's Product-moment Correlation Coefficient or r for short, is the most commonly used statistic to numerically describe a ____________, linear relationship between two variables
  - In this case, saying "non-directional" just means that it doesn't necessarily tell us about one predicting or ____________ another

> **Important**
>
> It is incredibly important to remember that r can only be applied when we think there is a *linear* relationship between two variables. A scatterplot can help identify if this does appear to be a linear relationship

4.2 Calculation

$$r = \frac{n\sum(xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

- Where: n: the number of ____________ data points x: individual data points in variable X y: individual data points in variable Y

> **Important**
>
> Like with everything in statistics, there are several ways to write this same formula - but this is the most simplified version.

- We'll come back to work though this equation in the Full Worked Example

4.3 Interpretation

- Value:
  - Always true: $-1 \leq r \leq 1$
  - $r = 0$ suggests ____________ relationship
  - $r = 1$ or $r = -1$ suggest a ____________ relationship, i.e., a straight line on a scatterplot

  - r closer to -1 or 1 (away from 0) suggests a ________ linear relationship
- Sign:
  - A ________ sign for r suggests a negative relationship
  - A ________ sign for r suggests a positive relationship
  - In relation to regression (which we'll discuss in a moment), r will have the same sign a b, or the slope of our line-of-best-fit
- Example:
  - I find that there is a strong positive correlation of GPA and SAT Score of $r = 0.76$
  - I find that there is weak negative correlation between salary and happiness (assume continuous) of $r = -0.21$
  - I find no relationship between mileage on cars and cost of cars of $r = 0.02$

> Discuss: Describe the relationship between two variables with r = -0.80

4.4 Significance

- r is a ________ statistic, and it's corresponding ________ in the population is represented as ρ (greek letter "rho")
  - We can ________ test to see if ρ is significantly different than 0

> **Important**
>
> Confusingly, there is a similar correlation coefficient for ordinal data called Spearman's rho, but that is not the same as what we are talking about here.

? If we are testing our continuous, numeric coefficient against the arbitrary standard of 0, what test does that sound most similar to, of the ones we have covered so far?

- (•) A) Welch's t-test
- ( ) B) One-sample t-test
- ( ) C) Chi-square goodness-of-fit
- ( ) D) Chi-square test-of-independence

Explanation:

- Our hypothesis pair:
  - $H_0 : \rho = 0$
  - $H_A : \rho \neq 0$
  - We decided whether we can ________ the null hypothesis that $\rho = 0$
- There are several ways to calculate the p-value (which we won't do by hand), but one of the common ways is via the ________!

4.5 Coefficient of Determination (r-squared)

- We can also get r^2, called r-squared or the coefficient of determination
  - This is a way of representing how much variance in one variable is ________ or accounted for by change in the other
- Example:
  - GRE scores correlate with later happiness, $r = 0.50 \rightarrow r^2 = 0.25$, thus 25% of the variance in happiness is explained by GRE score

Discuss: If I say that annual household spending and child GPA are correlated r = 0.12, write out how much variance is explained in the same manner as the above example.

4.6 Full Worked Example

- Consider the following small sets of data:
  - $X : 1, 2, 3, 4 \rightarrow \bar{x} = 2.5, s_x = 1.29$
  - $Y : 2, 3, 6, 3 \rightarrow \bar{y} = 3.5, s_y = 1.73$
  - $n = 4$ sets of data points

$$r = \frac{n\sum(xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

i	X	Y	X·Y	X^2	Y^2
1	1	2	2	1	4
2	2	3	6	4	9
3	3	6	18	9	36
4	4	3	12	16	9
Σ	**10**	**14**	**38**	**30**	**58**

$$r = \frac{4(38) - (10)(14)}{\sqrt{[4(30) - 10^2] * [4(58) - 14^2]}} = 0.4472$$

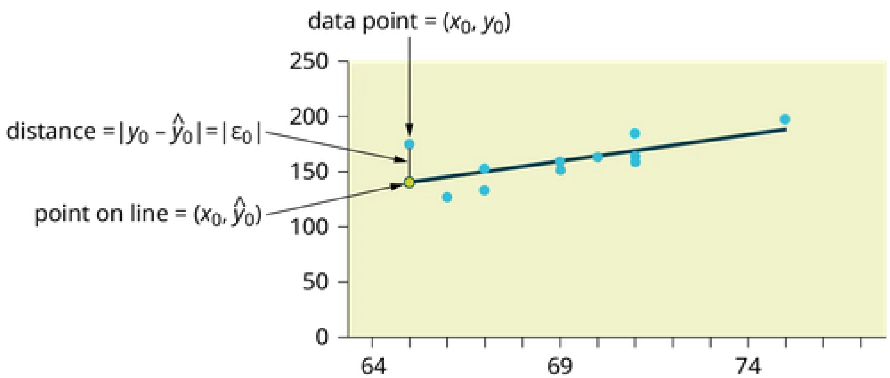
- $r = 0.4472$ is
  - A moderately strong, positive correlation coefficient
  - $r = 0.4472 \rightarrow r^2 = 0.20 \rightarrow$ 20% of variance in Y is accounted for by X

5 Regression Equation

5.1 Introduction

- While graphical methods are useful, we often want ________ to back up our conclusions
- The purpose of simple linear ________ analysis is to “fit” or “draw” a linear, straight line that goes through the center of all of our points, sometimes called the line-of-best-fit
  - There are ________ methods for the exact equation is used to minimize the distance to each point from our line-of-best-fit or error
  - For our simple use, the most common method is the ordinary least-squares

5.2 Minimizing Error



- Our line-of-best-fit is made up of ________ many predicted or expected points with coordinates: $(\hat{x}, \hat{y})$ and passes through our observed data with coordinates: (x, y)
  - Thus, our error is the distance between the observed and the predicted points: $|y - \hat{y}| = \epsilon$, put another way, the ________ on the y-axis that the predicted line is away from the observed point
  - We get an ϵ or error for each observed point:

$$(\epsilon_1)^2 + (\epsilon_2)^2 + ... + (\epsilon_n)^2 = \sum_{i=1} \epsilon^2 = SSE$$

- The above equation gives us the sum of squared errors (SSE)
  - In least-squares regression, our goal is to ________ a line that ________ the SSE

5.3 The Regression Equation Calculation

- The line-of-best-fit that is fit to ________ SSE can be found in the sample, using this formula:

$$\hat{y} = a + bx$$

- Where:
  - $\hat{y}$: our ________ y-value at any given point
  - $a = \bar{y} - b\bar{x}$: ________
  - $b = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum x - \bar{x}^2}$ or $b = r(\frac{s_y}{s_x})$: line ________
  - $\bar{x}$: Sample ________ of variable X
  - $\bar{y}$: Sample mean of variable Y
  - s_x: Sample standard ________ of variable X

- s_y: Sample standard deviation of variable Y
- r: Pearson's ____________ coefficient

- As you can see, there are multiple steps and ____________, but our hope is to get to a point in which we have values for both a and b

5.4 Worked Example

- Consider the following small sets of data:
  - $X : 1, 2, 3, 4 \rightarrow \bar{x} = 2.5, s = 1.29$
  - $Y : 2, 3, 6, 3 \rightarrow \bar{y} = 3.5, s = 1.73$
  - $n = 4$ sets of data points
  - $r = 0.4472$
- Regression math
  - $a = 3.5 - 0.60(2.5) = 2$
  - $b = 0.4472(\frac{1.73}{1.29}) = 0.60$-ish
  - Final line equation: $\hat{Y} = 2.0 + 0.60x$
  - Practical reading: For every one unit increase in X, there is a 0.60 unit increase in Y; at x = 0, y = 2.0
- Predicting Y off specific value of X
  - If $x = 2$ then...
  - $\hat{Y} = 2.0 + 0.60(2) = 3.2$

6 Conclusion

6.1 Recap

- Correlation and regression both give us some helpful ways of calculating information about the relationships between two continuous variables (bivariate)

- One can determine both the direction and relative strength of the relationships between the variables

- One can also make a determination on predicting on variable based upon another, with the caveat that there can always be the chance of error in that prediction

Key Terms

bivariate A description of data that is two variables 2

coefficient of determination r-squared, the percent of variable y variance explained by variable x 12
criterion A description of a variable that is treated as an outcome in a regression model 2

direction In correlation, the trend of how on variable reacts to change in another 7

error The distance between the predicted value and the observed value 13

line-of-best-fit In regression, the imaginary line drawn that minimize errors across the points 13

multivariate A description of data that is more than two variables 2

negative relationship In correlation, a trend where, as one variable increases, the other decreases 8

ordinary least-squares The mostly commonly used method to calculate error from each point from a regression line 13

positive relationship In correlation, a trend where, as one variable increases, so does the other 8
predictor A description of a variable that suggests that it is used as one of the data points to predict an outcome in a regression model 2

scatterplot A graphical way to represent two numeric variable with one variable on the x-axis and one variable on the y-axis 6
strength In correlation, how strongly related two variables are 7
sum of squared errors (SSE) In regression, the total sum of the of all of squared errors resulting from differences in the predicted and observed values 14